

Xiao Huang

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SUMMARY

Ph.D. Candidate at Duke University bridging software engineering and AI for Science research. Proven track record at **Google** architecting data analyzing pipelines over **PB-scale**. 20+ years pianist & 5y stock/option/future trader.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, TypeScript, Bash, R

Software Engineering: System Design, Distributed Systems, GCP, Data Pipelines, Git, Docker, CI/CD

Machine Learning & AI: Agent SKILL.md, PyTorch, HuggingFace, XGBoost, LLMs, RL, Anomaly Detection

Computational Chemistry Domain: MD Simulation, FEP, DFT, Quantum Mechanics, QM/MM, RDKit, AutoDock, QSAR

PROFESSIONAL EXPERIENCE

GCP Software Engineering Ph.D. Intern
Google LLC

May 2026 – Aug 2026
Kirkland, WA

- Architected a data aggregation pipeline for the **GCE metadata storage infra over Spanner**, achieving a **50,000x** size reduction to fuel a **SQL backend unsupervised Machine Learning (ML) Anomaly Detection Model**.
- Engineered **AI infrastructure** for Time-series Anomaly Detection by fine-tuning a **Bayesian Time-Series algorithm**, establishing robust models and metrics to monitor and describe clients' **behavioral metrics**.
- Developed a **full-stack architecture** featuring a **Python API layer** and **TypeScript frontend** to support various resolution queries and deliver a comprehensive interactive dashboard view.
- Pioneered a system that is **AI Agent native supportive** by developing a custom **SKILL.md**.
- **Productionized and deployed** all project components, now running reliably in production.
- Drove major business impact by reducing debugging time for on-call engineers solving outages and bug tickets, attracting **engineers** to learn and apply the pipeline to broader business logic.

AI for Science Research Intern
Deep Potential Technology

Feb 2022 – Aug 2022
Beijing, CN

- Developed **Covalent Uni-FEP**, a high-accuracy free energy perturbation (FEP) algorithm for covalent drug performance evaluation, benchmarking binding affinity errors within 1 kcal/mol (chemical accuracy).
- Automated complex inhibitor–protein topology setup and molecular dynamics (MD) preparation in **GROMACS** utilizing **Python** and **Bash**.
- Accelerated large-scale free energy computations by engineering a domain decomposition strategy with **multi-GPU support** and an **RDKit**-based reaction tracking pipeline.
- Authored core patent documentation and contributed production-quality code to the commercial software release.

SELECTED RESEARCH PROJECTS

De Novo Light-Driven Hole Bifurcating (HB) Proteins – AI-Driven Design
Computational Biophysics & Machine Learning

May 2023 – Present
Python, Bayesian Opt., XGBoost, DFT/MD

- Formulated multi-electron kinetic models and validated electron transfer driving forces via **DFT/MD/QMMM** simulations, mathematically deriving design rules to achieve near-100% HB reaction quantum yields.
- Processed and analyzed a 130k-point energetic landscape dataset using **XGBoost** and **SHAP** to extract and quantify critical **spatial and energetic design patterns** for optimal protein bifurcating ability.
- Integrated **SHAP-derived design rules with Bayesian Optimization** to efficiently explore structural and energetic design spaces, identifying optimal landscapes that **maximize light-to-biological energy conversion efficiency**.

LLM-Reinforcement Learning (LLM-RL) Pipeline for EGFR Drug Design
Generative AI & Drug Discovery

Feb 2025 – Aug 2025
Python, PyTorch, HuggingFace, AutoDock Vina

- Architected an end-to-end generative AI pipeline, fine-tuning a HuggingFace **ChemBERTa** Large Language Model via Masked Language Modeling for *de novo* small-molecule generation.
- Designed a custom **Reinforcement Learning (RL)** reward function balancing synthesizability, structural novelty, and **Random Forest**-predicted pIC₅₀ metrics to dynamically guide generation.
- Generated valid, novel drug scaffolds and rigorously evaluated performance through **molecular docking**, achieving a 30% improvement in predicted efficacy with > 60% structural uniqueness against baselines.

EDUCATION

Duke University

Ph.D. in Chemistry | *M.S. in Electrical and Computer Engineering (ECE)*

Durham, NC
Aug 2022 – May 2027

Emory University

B.S. in Chemistry | *Minor in Music* | **GPA: 3.81/4.0**

Atlanta, GA
Aug 2018 – Aug 2021

PUBLICATIONS

1. **Huang, X.**; DeGrado, W. F.; Therien, M. J.; Beratan, D. N. Designing multi-site charge-bifurcation networks in de novo proteins: a kinetic, statistical, and machine-learning approach. *Phys. Chem. Chem. Phys.* 2026, 28, 11161–11170. DOI: 10.1039/D6CP00473C
2. **Huang, X.**; Yuly, J.; Zhang, P.; DeGrado, W.; Therien, M.; Beratan, D. Design of Light Driven Hole Bifurcating Proteins. *ACS Cent. Sci.* 2025, 11, 10, 1911–1920. DOI: 10.1021/acscentsci.5c00803
3. Chai, Y.; **Huang, X (Corresponding Author)***. A Large Language Model-Guided Reinforcement Learning Framework for EGFR Anticancer Drug Design. *J. Comput. Aided Mol. Des.* DOI: 10.1007/s10822-025-00753-7
4. Smith, A. J.; Chen, C. D.; Wang, B.; **Huang, X.**; et al. From single step hops to emergent biological phenomena. *J. Inorg. Biochem.* 2026, 276. DOI: 10.1016/j.jinorgbio.2025.113191
5. Hruska, E.; Gale, A.; **Huang, X.**; Liu, F. AutoSolvate: A Toolkit for Automating Quantum Chemistry Design and Discovery of Solvated Molecules. *J. Chem. Phys.* 2022. 156, 124801. DOI: 10.1063/5.0084833